Haya Mahmoud

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Technical Skills

Languages: Java, Python, C, C#, JavaScript, TypeScript, SQL, HTML/CSS

Frameworks: React, Next.js, Node.js, Fastify, Vite, Tailwind CSS, Firebase

Cloud and Tools: GCP, BigQuery, Docker, AWS, Git, Linux/Unix, Bash, Jenkins, Jira, VS Code, Valgrind, Kubernetes

Concepts: Data Structures and Algorithms, OOP, REST APIs, Relational Databases, CI/CD, Unit Testing, Agile/Scrum

Professional Experience

Full Stack Software Developer Co-op *Toronto, ON / Remote*
FreshBooks *Sept 2026 – Present*

- Engineer and ship end-to-end full stack features, including an overtime pay feature in FreshBooks Payroll, contributing to weekly production releases on a financial platform that serves **30M+** users.
- Modernize **14+** REST API endpoints across **4+** payroll workflows, reducing average response latency by **8%** and improving the performance of third-party integrations.
- Re-architect **3** backend modules and **6+** integration components using **hexagonal architecture**, separating business logic from external dependencies to improve maintainability, scalability, and testability.
- Improve AI-assisted development efficiency by **15%** by optimizing **Claude Code** workflows with structured repository documentation, accelerating code exploration, feature development, and test generation.

Web Applications Developer Co-op *Mississauga, ON / Hybrid*
Kenna *May 2026 – Aug 2026*

- Developed and maintained full stack web applications for enterprise clients, including campaign sites, internal tools, and client-facing workflows supporting **20,000+** active users across the U.S. and Canada.
- Optimized features using modern technologies, improving performance by **30%** and strengthening security.
- Improved deployment reliability through code reviews and testing, reducing rollbacks by **15%**.

Projects

BugSpray *Nov 2026*
Python, Claude AI, Jira REST API, GitHub CLI, SQLite, Asyncio github.com/hayamahmoud/BugSpray

- Built an AI-powered bug-fixing tool integrating **Jira, Claude AI, and GitHub,** accelerating bug resolution by **24%** through automated code investigation, fix generation, and draft PR creation.
- Automated Jira backlog management across **3** workflow stages (sync, triage, resolution), enabling approve/reject decisions, blocking previously fixed bugs, and updating Jira to Done via GitHub **webhooks** after PR merge.
- Engineered sequential AI-agent queuing to process multiple approved bugs one at a time, preventing overlapping runs and reducing the risk of exhausting API credits mid-fix, with **4** safeguards for validation, duplicate prevention, cancellation, and human review.

Pray360 *Jan 2026*
TypeScript, React, MapTiler, Fastify, PostgreSQL, Docker github.com/hayamahmoud/Pray360

- Built a platform for discovering and reviewing prayer spaces using **React, Fastify, and PostgreSQL.**
- Engineered RESTful APIs with input validation and structured error handling, containerizing backend services with **Docker** for consistent deployment.
- Secured **3rd place** in the Open Innovation category at a competition with **300+ participants**.

ShelterMap Toronto *Nov 2025*
Python, Next.js, React-Leaflet, Google Cloud, BigQuery, Docker github.com/hayamahmoud/ShelterMap

- Engineered a Python data pipeline to process **500,000+** records, identifying high-priority shelter locations across the Greater Toronto Area through an interactive **Next.js/React** map.
- Leveraged **BigQuery** for large-scale data analysis and deployed containerized services using **Docker** and **GCP**.
- Earned a **Best Use of Google Cloud** nomination.

Education

Bachelor of Computing, Software Engineering (Co-op) *Sept 2024 – Expected Apr 2029*
University of Guelph

- Area of Emphasis: Artificial Intelligence

Awards & Achievements

University of Guelph's CEPS Dean's Honours List (4x) *Sep 2024 – May 2026*

- Recognized for achieving an average of **80%+** per semester.